

Text Classification Pipeline Forensics

AIAA3102 Final Project, Topic A

2026-07-21

Abstract

This project investigates a CPU-only classifier for the Disaster Tweets binary classification task, where target 1 denotes a tweet describing a real disaster. The objective is not simply to maximize target-1 F_1 : it is to determine which pipeline changes have credible evidence, which errors they exchange, and which conclusions remain uncertain. All learned transformations are fitted on training data only; each decision is selected on dev, frozen, and then evaluated once on held-out data. The final dev-selected operating point is balanced Logistic Regression with $C = 10$ and threshold 0.56. It reaches held-out F_1 0.754717, compared with 0.731984 for the required baseline. The reported gain is accompanied by explicit false-positive/false-negative trade-offs, surface-reliance stress tests, and a conservative data-quality audit that does not alter labels.

1 Project Problem and Goal

The project studies binary classification on the public Kaggle Disaster Tweets training data mirrored by UC Berkeley RISE [2]. A tweet with target 1 reports a disaster or related real-world event; target 0 does not. The supplied split is fixed and is never regenerated: 4,567 training rows, 1,523 dev rows, and 1,523 held-out rows. The corresponding positive counts are 1,962, 655, and 654.

The central question is not whether a different score can be produced, but whether a score change has a defensible mechanism. The project therefore asks whether the required TF-IDF plus Logistic Regression baseline reproduces the course reference; whether normalization and feature-source changes reduce surface dependence rather than exploit shortcuts; which decision rule gives the strongest dev-selected precision-recall operating point; and which duplicate or error evidence warrants an audit action rather than a label change.

The course contract reports reference F_1 0.757422. The submitted baseline reaches held-out F_1 0.731984, a gap of 0.025438. This gap is treated as a bounded forensic diagnosis rather than an invitation to choose whichever configuration is closest on held-out data.

2 Methodology

2.1 Split, fitting, and freeze protocol

Every vectorizer, categorical encoder, numeric scaler, and classifier is fitted only on the training partition. A dev command writes a pending decision; `freeze-ticket` copies that decision into `experiments/decisions.json`; only a frozen decision can run against held-out rows. `run-all` is the deterministic final regeneration command. Thus held-out data evaluates frozen configurations rather than choosing them.

Stage	Rows available	Permitted action	Persisted evidence
Train	4,567	Fit text and metadata transformations; fit model parameters	Fitted-ID checks and model configuration
Dev	1,523	Compare predefined candidates; select threshold, feature set, or model	Pending decision, dev predictions, candidate artifacts
Freeze	0 additional	Promote one dev choice without re-fitting	<code>experiments/decisions.json</code>
Held-out	1,523	Score the frozen configuration once; inspect transitions	Held-out predictions, summary, confusion counts

Table 1: Evaluation protocol.

All reported values are target-1 precision, recall, F_1 , and accuracy. For Tickets 2–5, `fixed_fp`, `fixed_fn`, `new_fp`, and `new_fn` are calculated against the same frozen Ticket 1 baseline, rather than against the immediately preceding ticket. This makes error movement comparable across the project.

The verified environment is Python 3.12.9 with NumPy 2.5.1 [1], pandas 3.0.3 [3], SciPy 1.18.0 [5], scikit-learn 1.9.0 [4], and Matplotlib 3.11.0. The seed is 3102. The full environment is recorded in `results/environment.json`, while the final artifacts are reconstructed from prediction files by the validator.

2.2 Ticket design

Ticket	Hypothesis and controlled lever	Dev selection evidence	Frozen held-out role
1	A baseline mismatch is caused by unspecified implementation details; vary one factor per diagnostic probe.	Fixed 26-probe matrix; baseline itself remains assignment-driven.	Forensic dev/held-out delta agreement only; probes cannot replace the baseline.
2	Stable URL replacement removes opaque URL-fragment reliance while retaining link presence.	Twelve one-lever normalization candidates.	Score the selected <code>url_replace</code> model and inspect transitions.
3	Keyword, location, and shallow counts may be useful but brittle shortcut signals.	Seven feature-source configurations and metadata interventions.	Score the retained text-only model.
4	A limited choice of regularization, class weighting, threshold, or LinearSVC can improve the operating point.	Fourteen Logistic Regression configurations plus one LinearSVC.	Score the frozen best dev configuration.
5	Duplicates, conflicts, and confident model errors expose audit evidence but do not automatically prove a label error.	No label-correction experiment is selected.	Audit the frozen Ticket 4 model without changing labels or rows.

Table 2: Controlled investigation design.

3 Main Evidence and Results

3.1 Ticket 1: Baseline discrepancy diagnosis

The frozen baseline uses lowercased word (1,2)-gram TF-IDF with sublinear term frequency, IDF and smooth IDF, a 40,000-feature cap, per-document L2 normalization, and `liblinear` Logistic Regression with L2 penalty, $C = 1$, seed 3102, and threshold 0.5. It obtains dev F_1 0.734034 and held-out F_1 0.731984. Its held-out precision is 0.777969, recall is 0.691131, and accuracy is 0.782666.

The diagnostic matrix fixes every unnamed variable, including the split, training-only fitting, threshold, and baseline parameters. It changes exactly one named factor at a time. The full 26-probe artifact includes negative controls as well as improvements. It was fixed before diagnostic held-out scoring and is explicitly not a held-out model-selection table.

One-factor probe	Dev F_1	Held-out F_1	Dev delta	Held-out delta	Interpretation boundary
Submitted baseline	0.734034	0.731984	0.000000	0.000000	Required comparison anchor
$C = 10$	0.750982	0.752137	+0.016948	+0.020153	Larger C weakens L2 regularization
$C = 4$	0.745067	0.756078	+0.011033	+0.024095	Diagnostic regularization response
Unigrams only	0.738812	0.751220	+0.004778	+0.019236	Removes phrase features; not selected
<code>min_df=3</code>	0.741176	0.750419	+0.007143	+0.018435	Removes many rare terms
<code>norm=None</code>	0.738333	0.758621	+0.004299	+0.026637	Preserves TF-IDF vector magnitude
<code>lbfgs</code> solver	0.734034	0.731826	+0.000000	-0.000158	Solver alone does not explain the gap
No IDF	0.720949	0.723370	-0.013085	-0.008613	Negative control
Preserve case	0.717619	0.709106	-0.016415	-0.022878	Negative control
L1 normalization	0.316176	0.288582	-0.417857	-0.443402	Strong negative control

Table 3: Representative rows from the fixed 26-probe baseline diagnostic matrix.

The 26 dev/held-out F_1 deltas have Pearson $r = 0.996813$ and Spearman $\rho = 0.874915$. Figure 1 visualizes this descriptive agreement. Its left panel expands near-zero deltas so that modest positive changes can be inspected; the right panel retains the full range, including the severe L1-normalization decline. The probes share the same data, baseline, and model family, so this association is not treated as independent-sample inference, a p-value claim, or proof of causality.

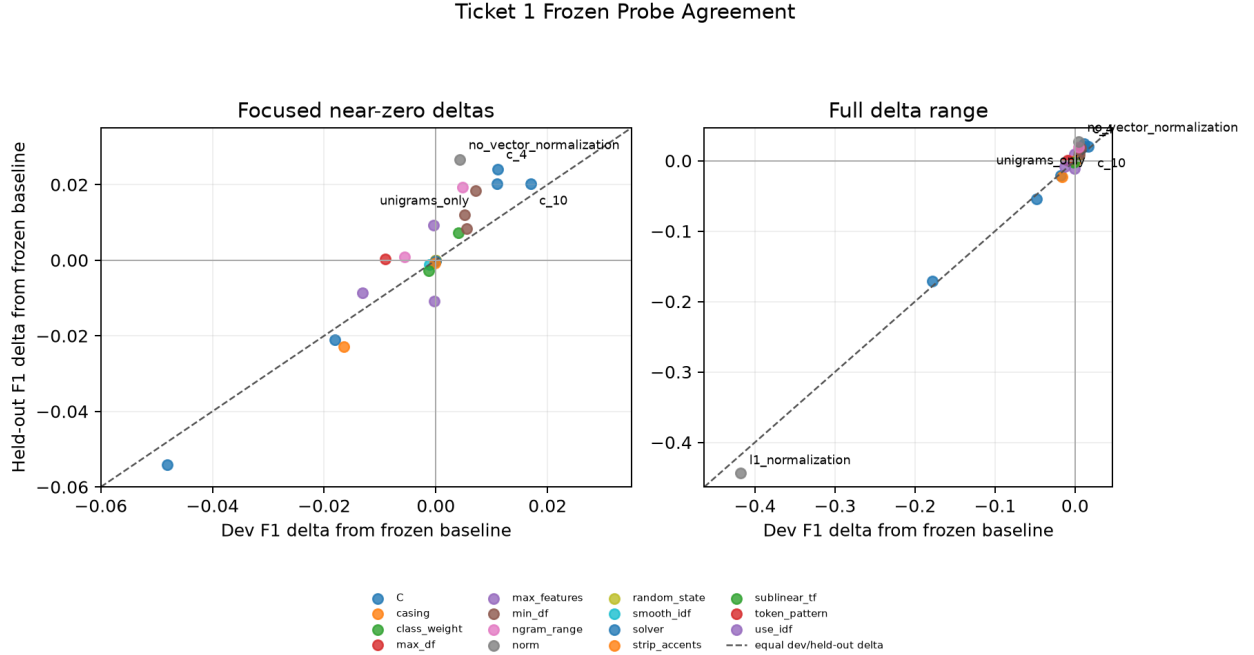


Figure 1: Ticket 1 fixed-probe dev/held-out delta agreement. The left panel magnifies near-zero changes; the right panel preserves the full negative range.

Two representative probes were designated after dev comparison and before held-out scoring to make aggregate F_1 movement inspectable at the row level.

Probe	Fixed FP	Fixed FN	New FP	New FN	What the counts show
$C = 10$	12	38	32	6	Recall-oriented movement: many recovered positives but a substantial FP cost.
<code>norm=None</code>	53	36	26	26	Broad movement in both directions; not a free improvement.

Table 4: Post-freeze representative probe transitions.

For example, $C = 10$ recovers ID 244, a real tweet about a shooting or airplane accident, while newly flagging ID 117’s conversational travel-delay wording. `norm=None` correctly rejects ID 303’s non-event “annihilated” promotional wording but newly misses ID 302’s terse positive “Annihilated Abs” text. The stable-ID rows are retained in `results/discrepancy_error_transitions.csv`; these examples illustrate prediction movement, not label corrections.

The baseline-reference gap remains unresolved because the reference vectorizer, classifier parameters, package versions, reference dev score, and reference predictions are unavailable. The evidence bounds plausible implementation causes but cannot identify a unique hidden reference configuration.

3.2 Ticket 2: Text normalization lever

Ticket 2 holds the Ticket 1 model, text features, split, and threshold fixed while changing one normalization decision at a time. The hypothesis is that a URL placeholder preserves the existence of a link without allowing opaque URL fragments to become fragile lexical features. No combinations of normalization changes are searched.

Candidate	Dev precision	Dev recall	Dev F_1	Delta from raw	Selected?
url_replace	0.810219	0.677863	0.738155	+0.004121	Yes
hashtag_remove	0.775338	0.700763	0.736167	+0.002133	No
url_remove	0.833333	0.656489	0.734415	+0.000381	No
Raw baseline	0.780069	0.693130	0.734034	+0.000000	No
mention_replace	0.780656	0.690076	0.732577	-0.001457	No
mention_remove	0.774744	0.693130	0.731668	-0.002366	No
preserve_case	0.758503	0.680916	0.717619	-0.016415	No

Table 5: Selected Ticket 2 dev candidates. Emoji replacement/removal, hashtag stripping, and punctuation removal/repetition exactly reproduce raw dev F_1 .

Figure 2 shows all 12 candidates as a scaled dot plot rather than a zero-baseline bar chart, so the small but real dev differences are visible without exaggerating bar lengths. Its stress panel compares the raw baseline and the frozen URL-replacement model under the same synthetic text perturbations.

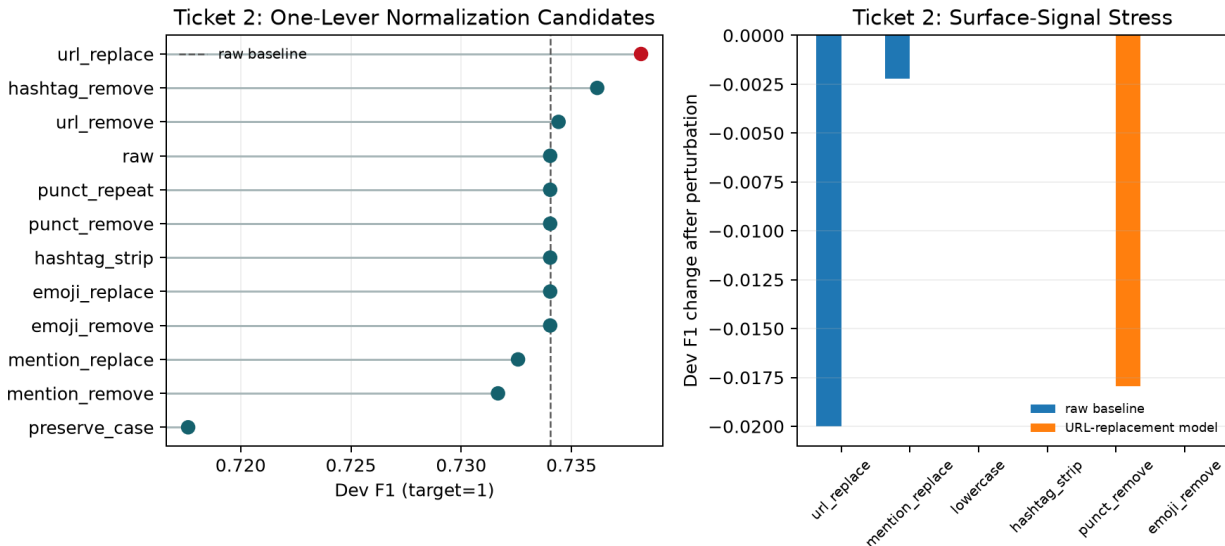


Figure 2: Ticket 2 normalization candidates and surface-signal stress results on dev. Red denotes the frozen URL-replacement choice.

After the dev decision was frozen, **url_replace** reached held-out F_1 0.739817 and accuracy 0.794485 (TN=765, FP=104, FN=209, TP=445). Relative to Ticket 1, it fixes 32 FPs and 8 FNs while introducing 7 FPs and 15 FNs. ID 353 is promotional “World Annihilation” content that becomes a fixed FP after URL fragments no longer dominate the score. ID 237 is a real airplane-accident post that becomes a new FN, demonstrating the recall cost. The conclusion is limited: URL replacement removes demonstrated URL-fragment dependence and slightly improves this dev-selected model; it does not make the classifier generally surface-invariant.

3.3 Ticket 3: Feature and shortcut audit

Ticket 3 tests whether supplied metadata or shallow counts add trustworthy information beyond tweet text. All candidates share the split, seed, classifier family, and threshold. Numeric features are tweet length, counts of words, URLs, mentions, hashtags, punctuation, and digits, uppercase ratio, and repeated punctuation. The selected text-only model is therefore an affirmative decision to reject shortcut additions, not an absence of experiments.

The left panel of Figure 3 retains every dev candidate. The right panel only perturbs models that consume metadata, avoiding the misleading result that changing an unused field proves robustness.

Text-only therefore remains the frozen Ticket 3 choice and produces the baseline held-out F_1 0.731984. This does not say that **keyword** is invalid task information; it is supplied context and contains real benchmark signal. It says that adding it to text lowers dev F_1 here and that models using it show strong synthetic dependence. Candidate

Model evaluated on dev	Perturbation	F_1 before	F_1 after	Change	Flips	Reading
Raw baseline	Replace URLs	0.734034	0.714029	-0.020005	125	Raw model relies on opaque URL fragments.
URL-replacement model	Replace URLs	0.738155	0.738155	0.000000	0	Intended invariance is achieved.
URL-replacement model	Remove punctuation	0.738155	0.720212	-0.017943	70	The model remains sensitive to a different surface transformation.
Raw baseline	Replace mentions	0.734034	0.731826	-0.002208	3	Small residual mention sensitivity.

Table 6: Ticket 2 dev-only perturbation evidence.

Feature source	Dev precision	Dev recall	Dev F_1	Delta from text	Selected?
Text	0.780069	0.693130	0.734034	0.000000	Yes
Text + numeric	0.786972	0.682443	0.730989	-0.003045	No
Text + keyword	0.747126	0.694656	0.719937	-0.014097	No
Keyword + location	0.675039	0.656489	0.665635	-0.068399	No
Keyword only	0.679612	0.641221	0.659859	-0.074175	No
Numeric only	0.616327	0.461069	0.527511	-0.206523	No
Location only	0.579439	0.094656	0.162730	-0.571304	No

Table 7: Ticket 3 feature-source ablation on dev.

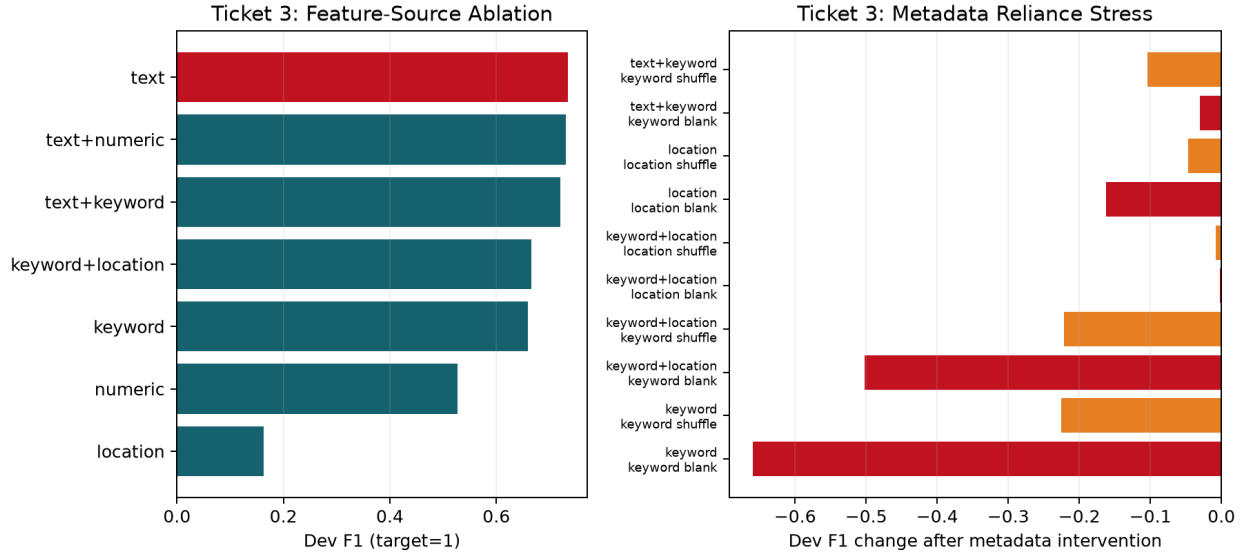


Figure 3: Ticket 3 feature-source ablation and metadata-reliance stress results on dev.

Consuming model	Intervention	F_1 before	F_1 after	Change	Flips	Evidence boundary
Keyword only	Blank keyword	0.659859	0.000000	-0.659859	618	Direct reliance on supplied keyword.
Keyword only	Shuffle keyword	0.659859	0.434231	-0.225627	699	Depends on row-to-keyword pairing.
Text + keyword	Blank keyword	0.719937	0.690009	-0.029928	247	Text model still changes when keyword is removed.
Text + keyword	Shuffle keyword	0.719937	0.616242	-0.103695	418	Stronger disruption from mismatched keyword metadata.
Location only	Blank location	0.162730	0.000000	-0.162730	107	Location is weak but entirely field-dependent.

Table 8: Ticket 3 dev-only metadata interventions.

transitions contain IDs 86 and 132 becoming fixed FPs and IDs 25 and 36 becoming new FPs; the exhaustive rows are in `results/error_transitions.csv`. Coefficients and perturbations are evidence of association and reliance, not causal semantics or deployment performance.

3.4 Ticket 4: Decision rule and model search

Ticket 4 keeps text features fixed and compares 14 Logistic Regression configurations—seven values of C times **None** or **balanced** class weights—plus one LinearSVC. Each Logistic Regression configuration chooses its threshold from 0.20 to 0.80 on dev in 0.01 increments. LinearSVC uses its decision function at zero. This is a deliberately limited CPU-only search, not an unrestricted hyperparameter sweep.

Figure 4 records the best dev F_1 for every Logistic Regression configuration. In this parameterization, larger C means weaker L2 regularization. The selected configuration is balanced LR with $C = 10$ and threshold 0.56. LinearSVC is retained as a genuine second-classifier comparison rather than omitted because it narrowly loses the dev competition.

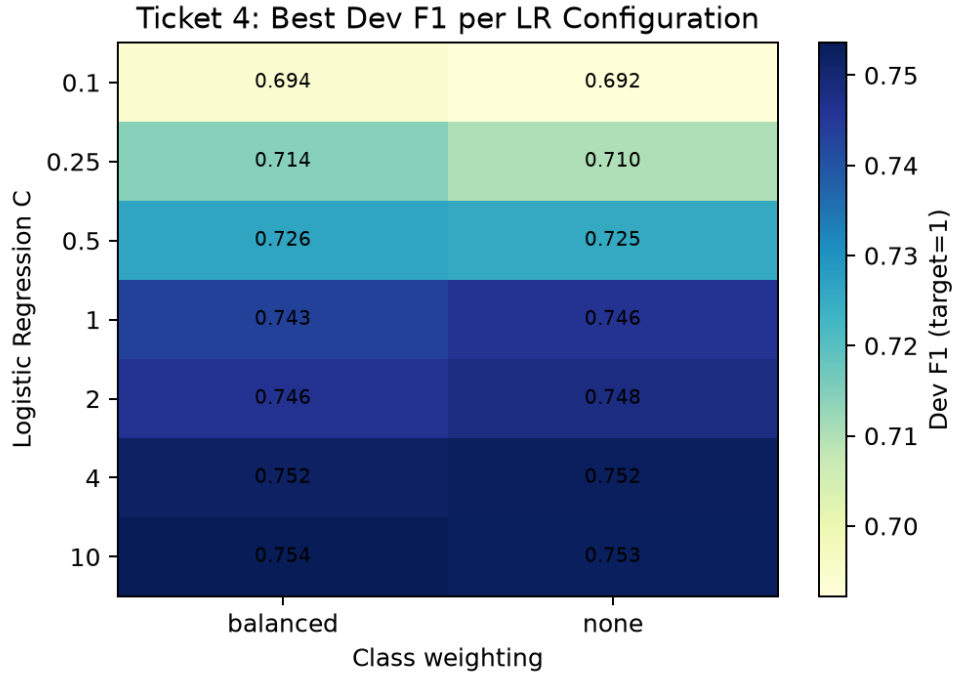


Figure 4: Ticket 4 dev F_1 for each Logistic Regression regularization and class-weight configuration.

Dev comparison	Threshold	Precision	Recall	F_1	What is isolated
Frozen Ticket 1 baseline	0.50	0.780069	0.693130	0.734034	Required baseline
Baseline threshold only	0.46	0.750000	0.741985	0.745971	Threshold effect alone
LR $C = 1$, balanced	0.51	0.741641	0.745038	0.743336	Class weight near baseline regularization
LR $C = 10$, unweighted	0.52	0.787980	0.720611	0.752791	Regularization with no class weighting
LinearSVC $C = 1$	0.00	0.772947	0.732824	0.752351	Second CPU classifier
LR $C = 10$, balanced	0.56	0.786070	0.723664	0.753577	Selected combination

Table 9: Dev ablation for the Ticket 4 decision.

The ablation distinguishes the selected combination from any single explanation. Threshold tuning alone improves dev F_1 by 0.011937, but does not reach the selected combination. The selected $C = 10$ balanced model trades some recall relative to lower thresholds for precision near the original baseline.

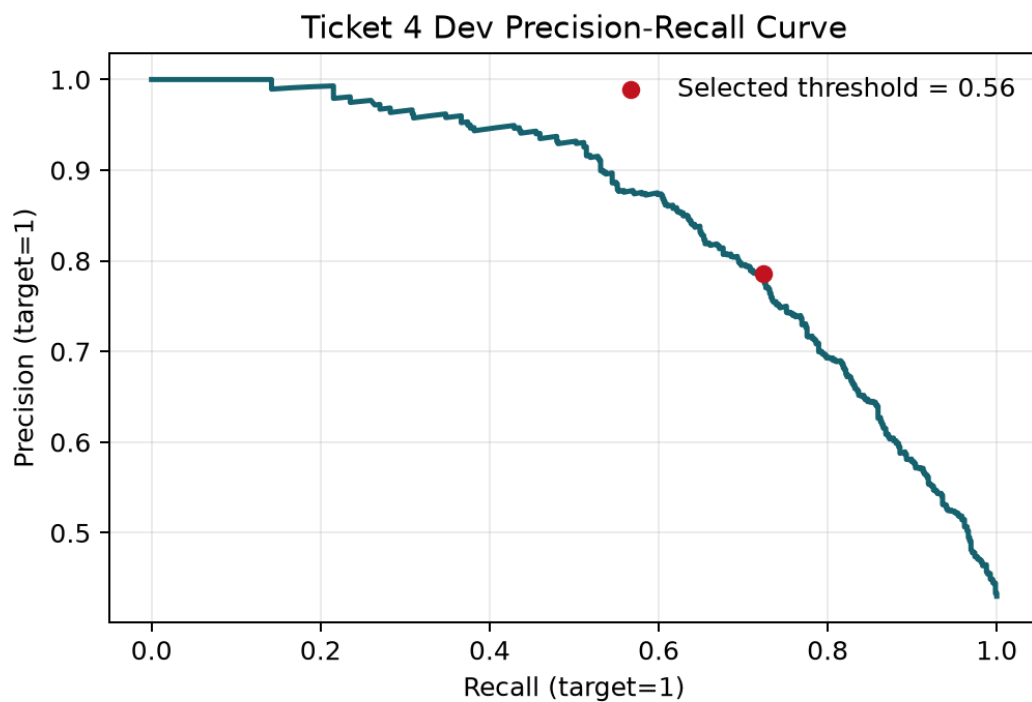


Figure 5: Ticket 4 dev precision-recall curve with the frozen operating point marked at threshold 0.56.

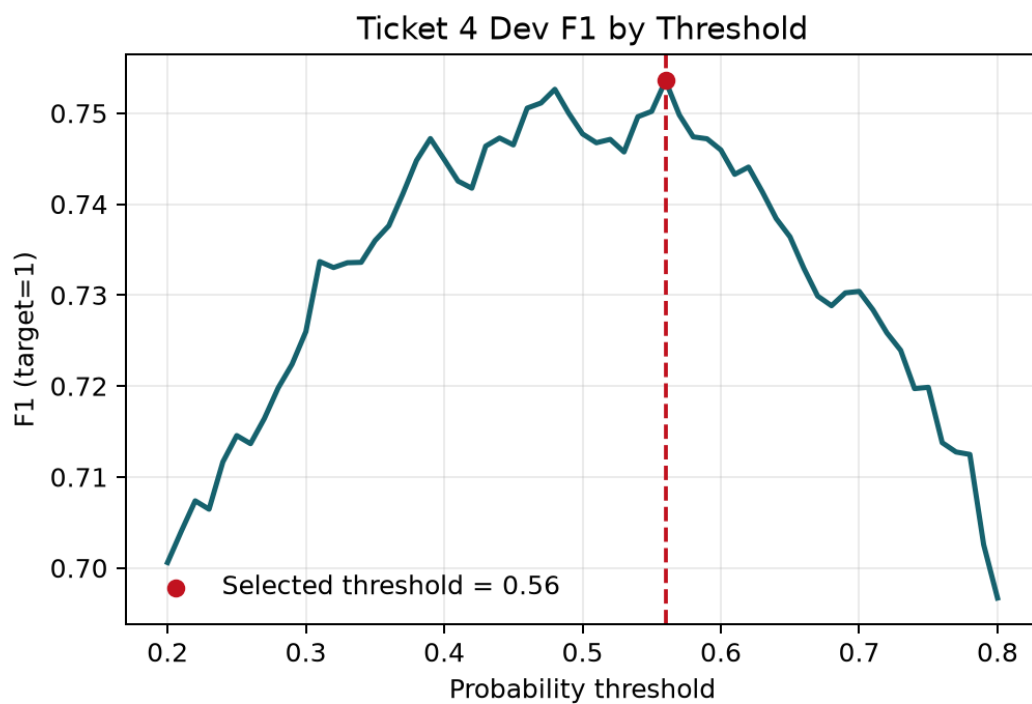


Figure 6: Ticket 4 dev F_1 across thresholds for the selected Logistic Regression configuration.

Figures 5 and 6 use dev scores only. They explain the selected operating point; they do not use held-out data to move the threshold after freezing. The frozen model reaches held-out F_1 0.754717, accuracy 0.795141, precision 0.776699, and recall 0.733945 (TN=731, FP=138, FN=174, TP=480). Relative to Ticket 1, it fixes 17 FPs and 37 FNs, while introducing 26 FPs and 9 FNs. The held-out result supports moderate agreement with the dev decision, but remains one evaluation of a selection made on a single dev split.

3.5 Ticket 5: Data-quality and error audit

Ticket 5 does not retrain a label-corrected model. It searches for exact duplicate text, normalized duplicate keys, character 3–5-gram TF–IDF near duplicates, conflicting labels inside duplicate groups, and high-confidence held-out errors from the frozen Ticket 4 model. A finding is classified as **keep_but_flag**, **ambiguous**, or **reject_false_positive**; no held-out label or row is modified.

Evidence type	Disposition	Evidence rows	Unique IDs	Held-out rows	Interpretation
Normalized duplicate	keep_but_flag	1,019	1,019	202	Repeated normalized text is a quality concern, not a label error by itself.
Conflicting-label duplicate	ambiguous	266	266	42	Duplicate conflict needs independent annotation.
Exact duplicate	keep_but_flag	179	179	41	Same-label duplicates are retained and flagged.
High-confidence FN	ambiguous	75	75	75	Model disagreement is not proof of a bad label.
High-confidence FP	ambiguous	48	48	48	Most apparent FPs remain review candidates.
Near duplicate	keep_but_flag	73	73	0	Consistent near duplicates indicate redundancy.
Near duplicate	ambiguous	27	27	0	Conflicting near duplicates need review.
High-confidence FP	reject_false_positive	1	1	1	Manual review supports the original label.

Table 10: Ticket 5 audit evidence summary.

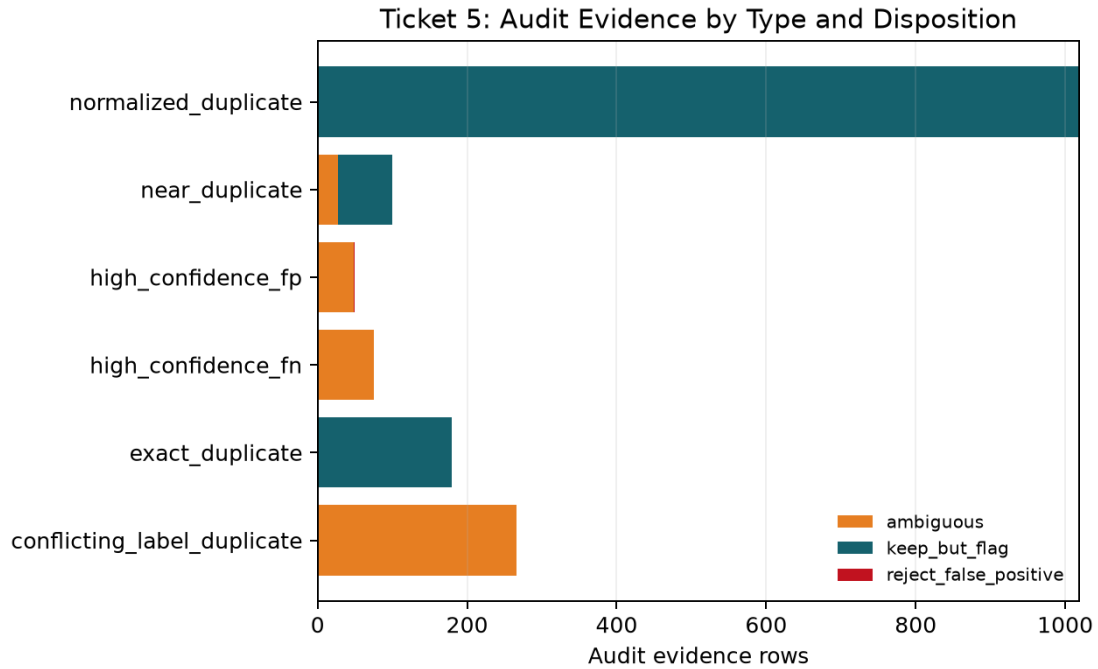


Figure 7: Ticket 5 audit evidence counts by issue type and disposition.

The audit has 1,688 evidence rows. IDs 59 (held-out) and 68 (train) are exact duplicate negative examples with cosine similarity 1.0, so both are **keep_but_flag**, not corrections. IDs 353 and 390 normalize to the same text

but have opposite labels, so the audit assigns **ambiguous** rather than guessing which annotation is wrong. ID 365 is a high-confidence FN with score 0.1080 and belongs to a conflicting duplicate group; it remains a review case. ID 198 is a high-confidence FP with score 0.8524: its text discusses lifetime odds of dying in an airplane accident rather than a reported event. Semantic review supports its original negative label, so the candidate model finding is deliberately rejected. The audit identifies reproducible risks and review priorities, but supplies no independent evidence strong enough to justify a label correction.

3.6 Cross-ticket held-out summary

Ticket	Frozen model or decision	Dev F_1	Held-out F_1	Accuracy	Fixed FP/FN	New FP/FN
1	TF-IDF (1, 2) LR	0.734034	0.731984	0.782666	0 / 0	0 / 0
2	URL replacement	0.738155	0.739817	0.794485	32 / 8	7 / 15
3	Text only	0.734034	0.731984	0.782666	0 / 0	0 / 0
4	Balanced LR, $C = 10$, $t = 0.56$	0.753577	0.754717	0.795141	17 / 37	26 / 9
5	Ticket 4 audit-only	0.753577	0.754717	0.795141	17 / 37	26 / 9

Table 11: Final frozen held-out results.

4 Case Analysis

The aggregate metrics conceal different kinds of model behavior. Table 12 connects representative stable IDs to their ticket-specific evidence and limits.

Table 12: Representative stable-ID cases.

Ticket	Stable IDs	Observation	Why the case matters	Boundary
1	244, 117	$C = 10$ recovers a real accident tweet but newly flags conversational accident wording.	F_1 gain contains both repaired FN and new FP movement.	A probe example is not a label correction or a new baseline.
1	303, 302	norm=None rejects non-event “annihilated” wording but misses terse positive wording.	Different preprocessing changes different semantic failure modes.	Two examples do not prove a global semantic mechanism.
2	353, 237	URL replacement fixes promotional “World Annihilation” content but misses a real airplane-accident post.	URL normalization reduces fragment reliance but can remove useful context.	The gain is small and selected on one dev split.
3	86, 132, 25, 36	Shortcut candidates create both fixed and new FPs.	Metadata-based changes are not uniformly beneficial at row level.	Exhaustive transitions, not these examples alone, support the decision.
4	244, 117, 2528	Final model recovers ID 244, creates an FP for ID 117, and misses traffic-collision ID 2528.	Higher F_1 still contains visible FP and FN costs.	F_1 is not a deployment-specific cost function.
5	59/68, 353/390, 198	Same-label duplicates are flagged; conflicting duplicates stay ambiguous; a manual review rejects a false label-error finding.	Audit disposition separates redundancy, ambiguity, and correction evidence.	No independent annotator was available.

The Ticket 4 operating point is preferable for the assignment’s target-1 F_1 only because 37 recovered FNs outweigh 9 newly missed positives under that metric. It also creates 26 new FPs while fixing 17. This is why the

report presents transitions, precision-recall curves, and examples alongside F_1 rather than treating the final score as a complete account.

5 Difficulties and Solutions

1. **The reference configuration was incomplete.** The contract provided a reference F_1 but not the vectorizer, classifier, package versions, reference dev score, or predictions. I froze the required baseline, tested a fixed one-factor probe matrix, and restricted held-out probe use to post-freeze forensic agreement rather than selecting the closest score.
2. **Preventing evaluation leakage required an executable protocol.** A narrative rule is insufficient if a CLI command can accidentally score held-out data. The pipeline uses pending dev decisions, an explicit freeze command, frozen decision records, train-fitted ID checks, and regression tests that reject held-out execution without a frozen decision.
3. **Aggregate F_1 did not explain error movement.** I added stable-ID prediction files, exhaustive transition categories, and artifact reconstruction. The validator recomputes F_1 , accuracy, and every transition count from predictions against the single Ticket 1 baseline.
4. **Initial shortcut perturbations could have been misleading.** Perturbing a metadata field on a text-only model would merely show that an unused field has no effect. The final stress design perturbs **keyword** and **location** only for models that consume them, making the observed changes interpretable.
5. **Data-quality evidence was easy to overclaim.** Duplicate text and high-confidence disagreement do not determine the correct label. The audit uses explicit dispositions and keeps every original label and held-out row, reserving `reject_false_positive` for the manually checked non-event ID 198.

6 AI Usage Declaration

AI assistance was used for code review, comparing the implementation with the handout, identifying missing experimental controls, proposing focused artifact exports, drafting and reorganizing documentation, and checking wording. It was treated as a proposal mechanism rather than a source of evidence. Every numerical claim was verified against regenerated CSV artifacts; quoted stable-ID examples were checked against the source data or audit records; the dev/freeze/held-out boundary was inspected in code; and the report was rebuilt locally. Final verification ran `validate-data`, `run-all`, `validate-artifacts`, and 60 automated tests successfully. A concise, non-verbatim work record is retained in `logs/chat.md`.

7 Discussion and Limitations

The project has several important limitations. First, a single dev split selects normalization, features, regularization, class weighting, classifier family, and threshold. Small dev differences may be selection noise, particularly after multiple comparisons. Held-out scores validate frozen decisions but do not eliminate this uncertainty.

Second, F_1 treats precision and recall symmetrically. Threshold 0.56 is an operating point optimized for the assignment metric, not a calibrated policy for a deployment setting with known false-positive and false-negative costs. The precision-recall and F_1 -threshold figures reveal the trade-off but cannot choose a real-world cost ratio.

Third, the Ticket 1 delta association is descriptive. The 26 probes are not independent experiments, and several settings alter related aspects of the same model. It supports reproducible directional behavior in this pipeline, but cannot prove a causal explanation of the undisclosed reference configuration.

Fourth, the URL and metadata perturbations are synthetic interventions. They establish sensitivity to transformations applied here, not robustness to all possible social-media shifts. Similarly, near-duplicate cosine thresholds are heuristic and high model confidence is not independent human annotation. Ticket 5 intentionally does not turn these cues into label changes.

8 Conclusion

The required baseline is deterministic and reproducible in the submitted environment, but it does not match the undocumented course reference configuration. The fixed Ticket 1 matrix narrows plausible sources of that gap

without allowing held-out diagnostics to redefine the baseline. URL replacement gives a small dev-selected improvement and removes a measurable reliance on opaque URL fragments, while remaining sensitive to punctuation changes. Keyword and location metadata provide some benchmark signal but are weaker than text and visibly brittle under the interventions used here.

The strongest final dev-selected configuration is balanced Logistic Regression with $C = 10$ and threshold 0.56, reaching held-out F_1 0.754717. Its gain is real in the fixed split, but it has explicit error costs and remains subject to selection uncertainty. The data-quality audit adds a final constraint: duplicated or ambiguous evidence should be reported and reviewed, not converted automatically into changed labels. Overall, the project supports careful, reproducible model forensics rather than score-only optimization.

9 References and Appendix

References

- [1] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
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- [3] Wes McKinney. Data structures for statistical computing in python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010. URL <https://pandas.pydata.org/>.
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A Artifact Map

Question	Primary machine-checkable evidence
Split and environment	<code>data/split_indices.json</code> , <code>results/environment.json</code>
Frozen decisions	<code>experiments/decisions.json</code> , <code>experiments/pending_decisions.json</code>
Ticket 1 probes	<code>results/discrepancy_comparison.csv</code> , <code>results/discrepancy_association.json</code> , <code>results/discrepancy_error_transitions.csv</code>
Ticket 2 candidates and stress	<code>results/ticket2_normalization_comparison.csv</code> , <code>results/perturbation_stress.csv</code>
Ticket 3 feature and stress evidence	<code>results/ticket3_feature_comparison.csv</code> , <code>results/top_features.csv</code> , <code>results/perturbation_stress.csv</code>
Ticket 4 grid and operating point	<code>results/ticket4_model_grid.csv</code> , <code>results/decision_ablation.csv</code> , <code>results/ticket4_dev_decision_curve.csv</code> , <code>results/ticket4_dev_precision_recall_curve.csv</code>
Ticket 5 dispositions	<code>results/data_quality_audit.csv</code> , <code>results/ticket5_audit_summary.csv</code>
Final reconstruction	<code>predictions/heldout_predictions.csv</code> , <code>results/summary.csv</code> , <code>results/error_transitions.csv</code>

B Reproduction

From the repository root, run:

```
conda run -n aiaa3102 python -m pytest -q
conda run -n aiaa3102 python -m pipeline.cli validate-data
conda run -n aiaa3102 python -m pipeline.cli run-all
conda run -n aiaa3102 python -m pipeline.cli validate-artifacts
cd report && latexmk -pdf report.tex
```

The first four commands regenerate and validate the machine-checkable artifacts. The final command compiles this report; the figures are referenced relative to the `report/` directory.